Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. We trained each model for 50 epochs with a batch size of 64.

2 Results

Figure 1 shows the accuracy by dataset.

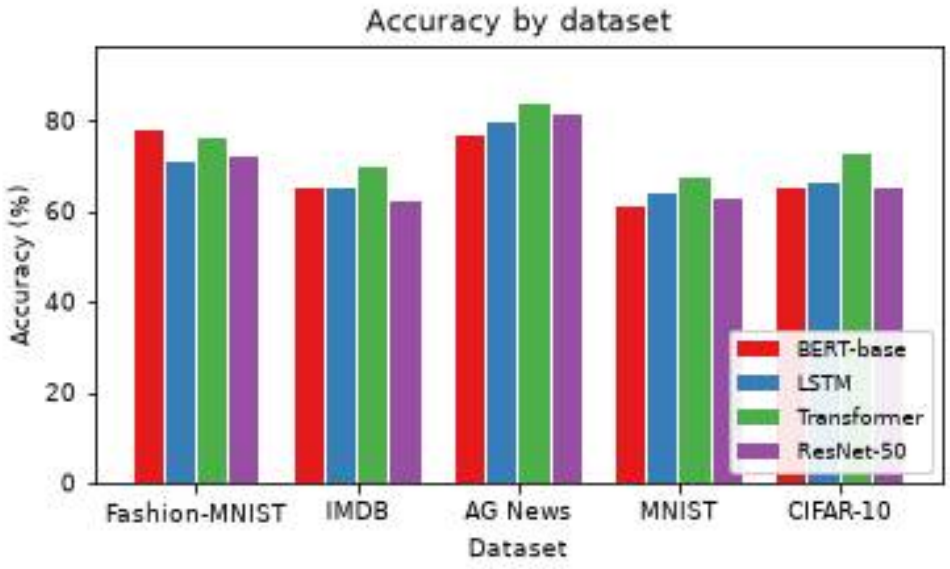


Figure 1: Accuracy by dataset for the models in this experiment.

On CIFAR-10, Transformer obtains a accuracy of 70.6%. BERT-base outperforms Transformer by 1 percentage points on Fashion-MNIST (Figure 1). On MNIST, Transformer is 25.8 points better than BERT-base.

Figure 2 shows the accuracy during training.

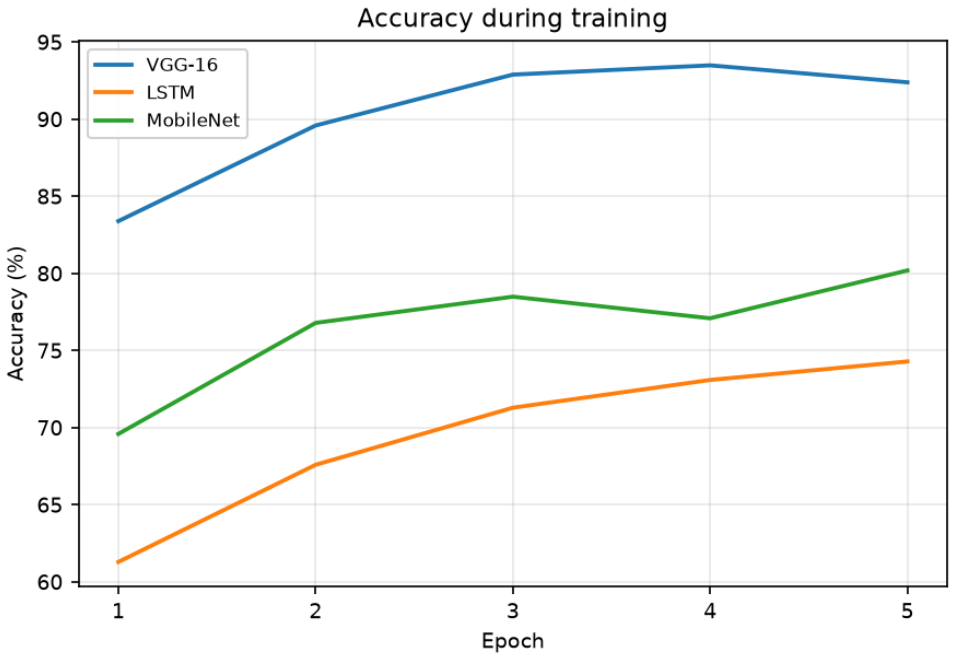


Figure 2: Accuracy during training for the models in this experiment.

At epoch 2, LSTM achieves about 70% accuracy. From epoch 4 to epoch 5, the accuracy of MobileNet

improves by a relative 13.2% (Figure 2).

Figure 3 shows the validation accuracy by dataset.

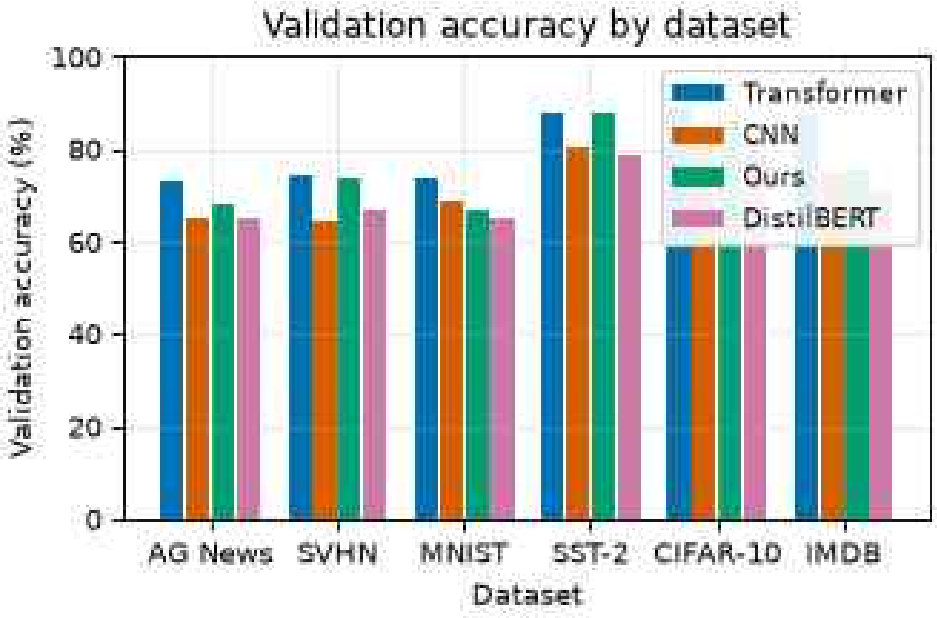


Figure 3: Validation accuracy by dataset for the models in this experiment.

On AG News, CNN obtains a validation accuracy of about 70%. Transformer achieves 68.6% validation accuracy on MNIST. CNN reaches 64.1% validation accuracy on SVHN. As shown in Figure 3, CNN reaches over 60% validation accuracy on SVHN.

Figure 4 shows the accuracy of CNN by dataset.

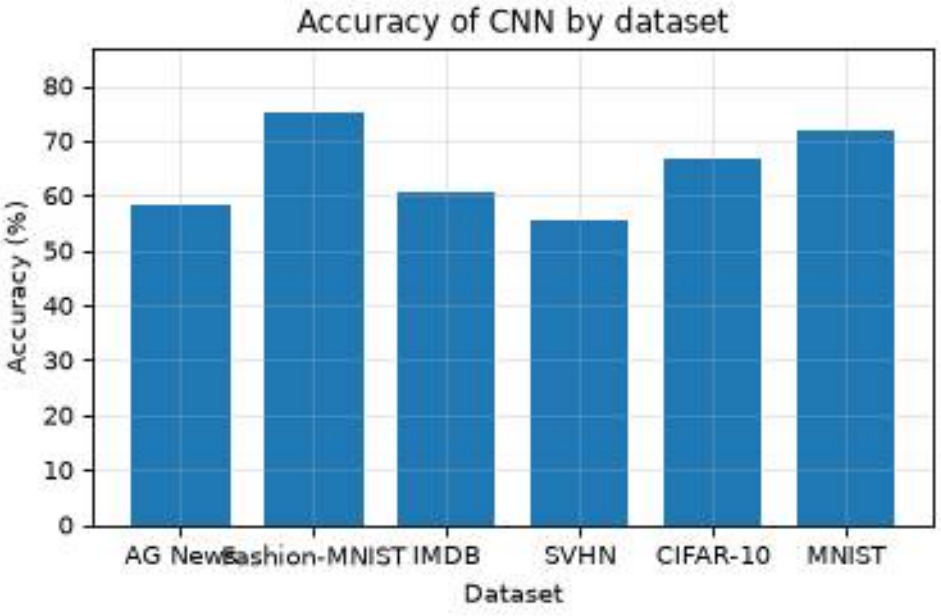


Figure 4: Accuracy of CNN by dataset for the models in this experiment.

On IMDB, CNN obtains a accuracy of 75.3% (Figure 4). As shown in Figure 4, CNN scores 2.7 points higher on CIFAR-10 than on SVHN. CNN reaches 38.0% accuracy on SVHN.

3 Discussion

Logistic Regression reaches 56.1% accuracy on CIFAR-10. We note that the training was stable for all models.